NET-METERING REFERANCE GUIDE FOR THE DISCOs



How the application process and the installation of Net Metering connection works

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Glossary

AEDB	Alternative Energy Development Board
CCE	Connection Charge Estimate
DG	Distributed Generator
DISCO	Distribution Company
EN	European Standard
IEC	International Electro-technical Commission
IEEE	Institute of Electrical and Electronics Engineers
KW	Kilowatts
kWh	Kilowatt hour
kWp	Kilowatt peak
MW	Megawatt
NEPRA	National Electric Power Regulatory Authority
NGO	Non-governmental organization
PKR	Pakistani Rupee
PV	Photovoltaic
RE	Renewable Energy
UL	Underwriters Laboratories

1. Introduction

This guide provides an overview of important points to consider when then consumer apply for the Net Metering as per NEPRA (Alternative and Renewable Energy) Distributed Generation and Net Metering Regulation 2015. This guide attempts to walk the DISCO office and DISCO's personnel through the different stages beginning from the application by the consumer up to the installation of Net Metering DG facility and the

1.A About Net Metering

Net metering is an electricity policy for consumers who own Renewable Energy facilities which allows them to use electricity whenever needed while getting credit for contributing their production to the grid.

Producing electricity partly for own consumption, and partly for sale to the DISCO, is now available in the Pakistan. Solar and Wind Energy is a long term power solution. The Solar PV Technology gives access to affordable electricity supply during system life. Residential and commercial customers can switch their electricity load to Solar/Wind energy and can slash their power bills.

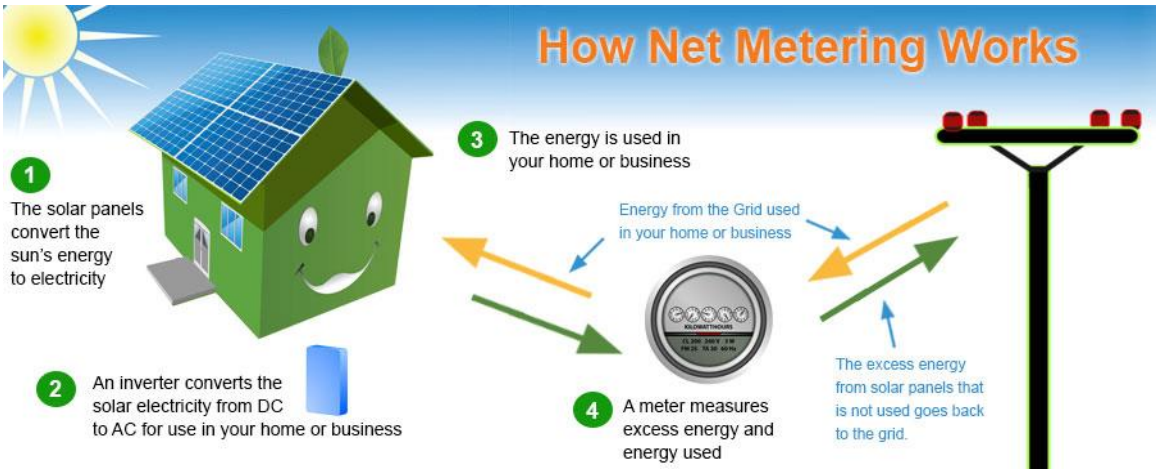


Figure 1.1 Net Metering process

This picture illustrates the flow of electricity from power generation via high voltage transmission and distribution utilities to the end-user who can now install a renewable energy facility and send the not needed electricity back to the distribution grid and earn credit for his export.

1.B Background

Pakistan has been facing energy crisis for the past few years as the demand and supply gap has widened. The country's current energy demand far exceeds its generation resources, and facing an energy shortfall of 4000 MW as a result load-shedding and power blackout have become severe issue.

Keeping in mind above issues The Government of Pakistan promotes investment in the generation of small scale distributed renewable energy, through the Alternative Energy Development Board (AEDB), on the basis of net-metering concept.

1.C Net Metering in Pakistan

The National Electric Regulatory Authority (NEPRA) announced the official Distributed Generation and Net Metering Regulations on September 1st, 2015. As per these regulations, any customer of the electric grid (three-phase connections) can avail the possibility of Net Metering for small-scale renewable energies installations.

In Pakistan, net-metering is the first policy mechanism of the Renewable Energy Act of 2006 which has been fully implemented. Section 8.4.2 of the Renewable Energy Act of 2006 provides that subject to technical considerations and without discrimination and upon request by distribution end-users, DISCOs shall enter into net-metering agreement with qualified end-users who will be installing the RE system.

NEPRA (Alternative & Renewable Energy) Distributed Generation and Net Metering Regulation 2015 is added in annexes

1.D Role of Distribution Code Review Panel (DCRP)

DCRP was established in October 21st 2014 for the integration RE plant to be connected with DISCOs. DCRP is a standing body to undertake the functions detailed in Distribution Code. The DISCOs are bound to follow the Distribution Code for all connections.

2. Steps involve in Net Metering

This chapter covers the detailed overview of the each step involved in the Net Metering procedure as per NEPRA's Regulation along with the timeline. Interconnection of consumer and DISCO and interconnection of DISCO and NEPRA covers in this section.

2.A Overview and timeline of the Procedure for Net Metering Connections

The Power Division, Ministry of Energy (MoE) is promoting net metering and intending to provide the services to the customers at their door step. Therefore, MoE has desired that the installers of systems for net metering (DG installers) as shortlisted by AEDB will be required to process the application for net metering on behalf of consumers applying for net metering connections up to 250 kW capacity. NEPRA has improved/shortened the timeperio for processing net metering applications. Moreover, MoE has asked DISCOs to get applications approved and connections energized within one month of receipt of application from the consumers.

The Flow of steps involve in Net Metering Process along with the timeline for net metering connections are as follows.

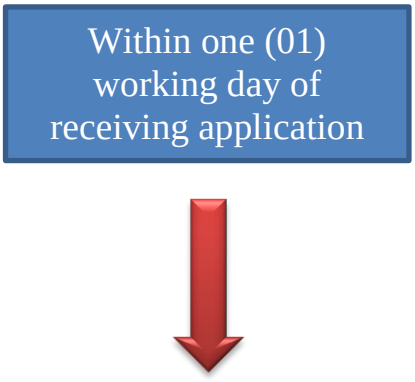
2.A.1 Application for Net Metering

Any applicant who meets the requirement of DG as defined in NEPRA's regulation, submits their application along with the necessary documents to the focal person of the DISCO. Application form is attached at the Annexure and can also be downloaded from NEPRA's website.



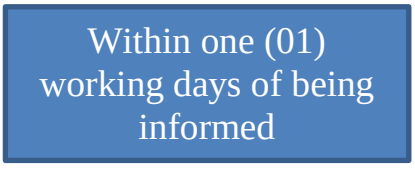
2.A.2 Acknowledgement of application

The DISCO will acknowledge its receipt and inform the applicant whether the application is completed in all respect or not. For the purpose of net metering connections up to 250 kW capacity, NEPRA has relaxed requirement of inspection from Electric Inspector.



Incomplete application form

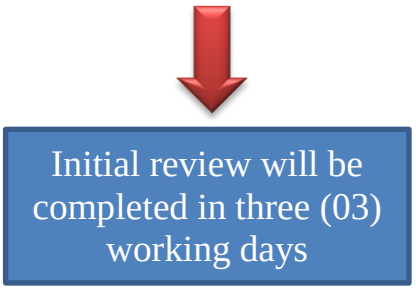
In case the application found incomplete, the DISCO will return the application the same day and will ask the applicant to complete the application and re-submit it. The applicant will have to submit the application in



complete form to the DISCO office.

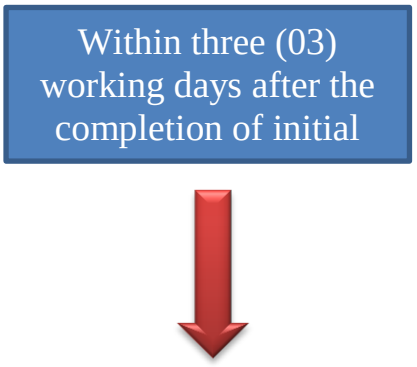
Initial review

Upon being satisfied that the applicant is complete in all respect, The DISCO office will perform an initial review to determine whether applicant qualifies for interconnection facility and required additional requirements, like comparison between sanctioned load and the size of the system installed.



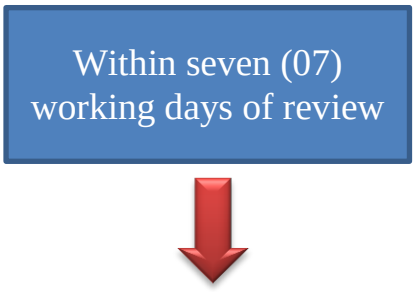
2.A.3 Technical feasibility in initial review

As per DISCOs, for connections up to 250 kW, no technical feasibility study is needed. Power Division, Ministry of Energy has directed the DISCOs to carry out technical studies and approve the connections at the Sub-Division level. In case the initial review reveals that the proposed facility is not technically feasible, the DISCO will return the application and communicate the reason to the applicant.



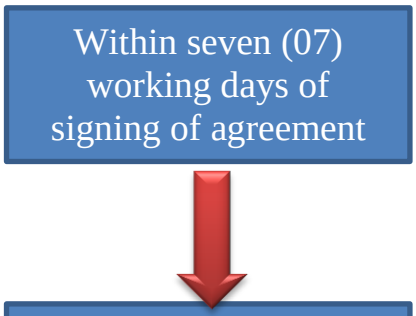
2.A.4 Agreement

If the DISCO office is satisfied that the applicant qualifies as a DG, then the DISCO and the applicant will enter into agreement. Agreement form (Schedule-I) is added in the annexure.



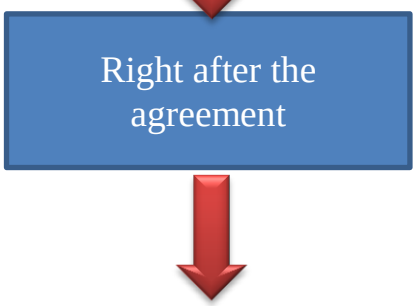
2.A.5 Generation license

The DISCO office will send the copy of the agreement between the applicant and the DISCO to the NEPRA along with the application for the issuance of Generation license. NEPRA will issue the Generation Licence within seven working days after receipt of requisite documents by DISCOs.



2.A.6 Connection Charge Estimate

After the agreement DISCO office will issue the Connection Charge Estimate (CCE), if any, to the applicant for the proposed interconnection facility up to the interconnection point including metering installation.



The applicant have the choice either it obtains the bi-

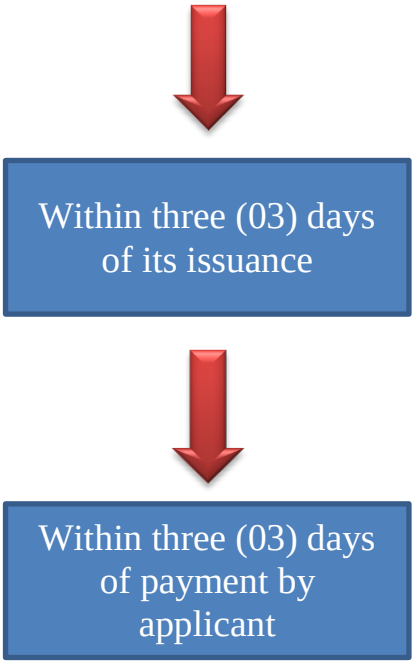
directional meter from DISCO or purchased it from private sector manufacturers and get it certified from the concerned DISCO as per technical requirements notified by NTDC.

2.A.7 Payment of CCE

If applicant choose to obtain the meter from concerned DISCO and DISCO has given the notice to the applicant to make the payment of Connection Charge Estimate (CCE), the applicant will pay the same and notify the DISCO office.

2.A.8 Installation of interconnection facility

When the charges will be paid the DISCO office will install and commission the proposed interconnection facility after the confirmation of Generation License to the DG by NEPRA.



3. Interconnection Requirements

This chapter will cover the requirements that are needed to be checked by the DISCO office in proposed DG facility by the consumer.

3.A Initial requirements

At the beginning of application procedure DISCO office has to make sure that these initial requirements are to be fulfilled by the applicant.

3.A.1 Technical feasibility requirement

After receiving an application for the interconnection facility, technical feasibility report is to be prepared by the inspection committee keeping in view the following checks:

a. The capacity / load on the transformer should be within the technical limits.
b. Whether the proposed interconnection would require upgrading the capacity of existing distribution network.
c. Ensure phase balancing to avoid unbalancing of the load in secondary circuit of distribution line.

3.A.2 NOC by Electrical Inspector

As per amendments proposed by MoE to NEPRA, the requirement of NOC by Electric Inspector for net metering connections up to 250 kW is omitted. NOC by Electrical inspector should be provided by those applicants along with their application form that are installing net metering connections above 250 kW.

Scope of the NOC includes grounding tests, insulation tests, wiring size suitability and general safety of complete internal wiring.

3.A.3 Grid Interconnection Standard: Electrical Parameters

The DG shall operate and inject power when the network is within the parameters mentioned hereafter. DG system shall disconnect itself from the utility network when the electrical parameter of the network falls out of the specified range.

3.A.3.1 Range of Operation: Voltage

The under voltage and over voltage levels and the corresponding triptimes shall be in accordance with IEC 61727**or better.**

Voltage	Maximum Trip Time(Seconds)
V < 50%	0.1
50% ≤ V < 85%	2.0
85% ≤ V < 110%	Continuous Operation
110% ≤ V < 135%	2.0
135% ≤ V	0.05

3.A.3.2 Range of Operation: Power Factor

The power factor must comply with requirements as per IEC 61727. The DG system shall have power factor greater than 0.9 when the output is greater than 50% of System rating. The Power Factor requirements set for respective consumer type by the utility company will be will remain applicable and larger DG (such as industrial connections) may be required to install additional systems for power factor correction.

3.A.3.3 Range of Operation: Frequency

The under frequency and over frequency levels and the corresponding trip times shall be as follows:

Frequency Range (Hz)	Maximum Trip Time (Seconds)
$f_{nominal} < 47.5$	0.1
$47.5 \leq f_{nominal} \leq 51.5$	Continuous Operation
$51.5 < f_{nominal}$	0.1

3.A.3.4 Harmonics

The inverter shall meet the Harmonic current injections in accordance with IEC 61727. The total harmonic distortion shall be less than 5% at 100% rated power of the inverter. Distortion limits for Even Harmonics and odd Harmonics are listed below:

Odd Harmonics	***Distortion Limit***
3rd through 9th	*Less than 4.0%*
11th through 15th	*Less than 2.0%*
17th through 21st	*Less than 1.5%*
23rd through 33rd	*Less than 0.6%*
33rd+	*Less than 0.3%*

Even Harmonics in these ranges shall be less than 25% of the odd harmonics listed. THD should be less than 5%

3.A.3.5 DC Current injection

The DC Current injection must be in accordance with IEC 61727. The PV system shall not inject DC current greater than 1% of the rated inverter output current into the utility interface.

3.A.3.6 Islanding Protection

Islanding protection shall be in accordance with IEC 61727. A PV system that ceases to energize the utility line in case of a voltage and frequency situation outside of the ranges stated in IEC 61727 within the timeframes set in the IEC 61727 is considered to be sufficiently protected against islanding.

3.A.3.7 Reconnection to the network

In case of utility line outage, fault or out of range, the inverter shall remain disconnected and

provide Islanding Protection. The inverter shall not energize the utility line for at least 1 minute after the service voltage and frequency have recovered to the continuous operations range parameters as defined above.

3.A.4 Inverter standards

As per Regulation by NEPRA, the grid connected inverters shall comply with UL 1741 standard which addresses the electrical interconnection design of various form of generating equipment. Other standards that the DG facility should adopt to are:

a. IEEE 1547 2OO3 (standard of interconnecting distributed resources with electric power system).
b. IEC 61215 for crystalline PV modules
c. IEC 61646 for thin film PV modules

3.A.5 Safety and EMC Standard

For system greater than 10kW, the inverter needs to comply with additional certifications and meet additional Electromagnetic compatibility and Safety Standards such as:

1) IEC EN 61000-6
2) UL1741
3) EN 62109-1/2
4) IEEE1547

3.A.6 Load flow study

For the Distributed Generators having an installed capacity of more than 500 kW, load flow study (on PSSE software) is compulsory and for the DG facility having capacity less then 500K load flow study can be achieved by FDRANA

Load flow study for the facility having capacity up to 250 kW is not required.

3.A.7 Upgrading the Transformer

In case if the load capacity due to already installed DG facilities at any transformer reaches 80%, the DISCO shall not grant approval for any new DG facility at that transformer unless the new DG Facility operator agrees and pay for the cost of augmenting the transformer capacity.

3.B Mandatory requirements

According to NEPRA Regulation, DISCO and applicant have to comply with some safety requirements as stated below:

3.B.1 Interconnection disconnect switch

DISCO personnel has to make sure that the applicant has installed an interconnection disconnect switch rated for its voltage and fault current requirements which will cut the flow of energy back to the grid if required. That disconnect switch should be locked and can be accessible by both DG and the DISCO personnel and shall meet the applicable IEC and IEEE